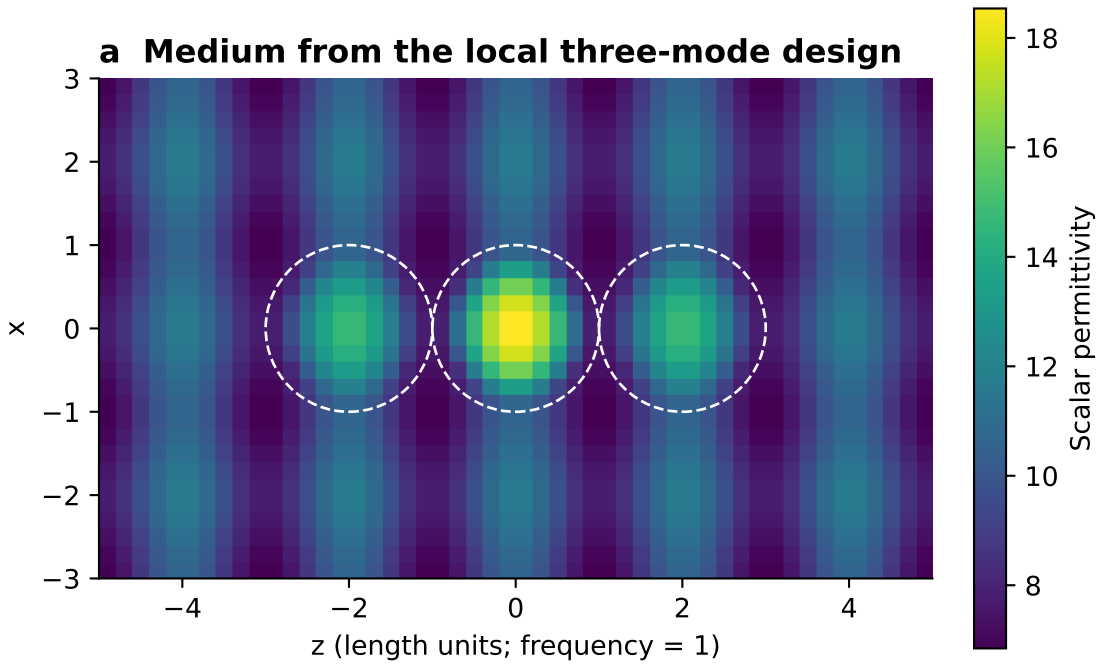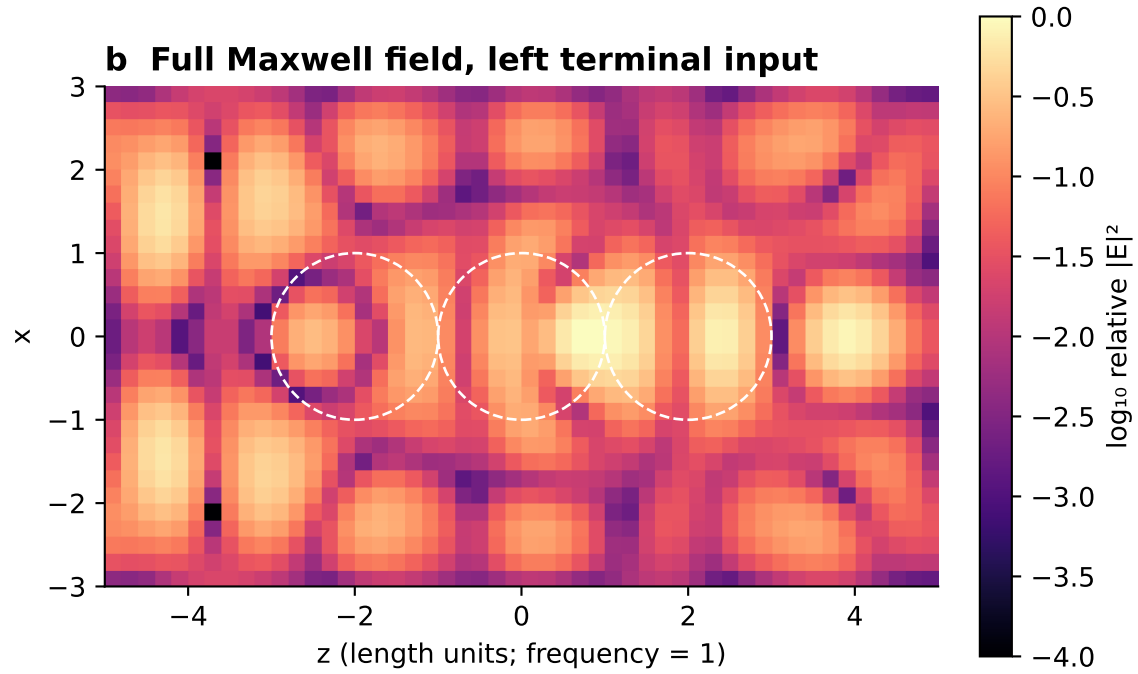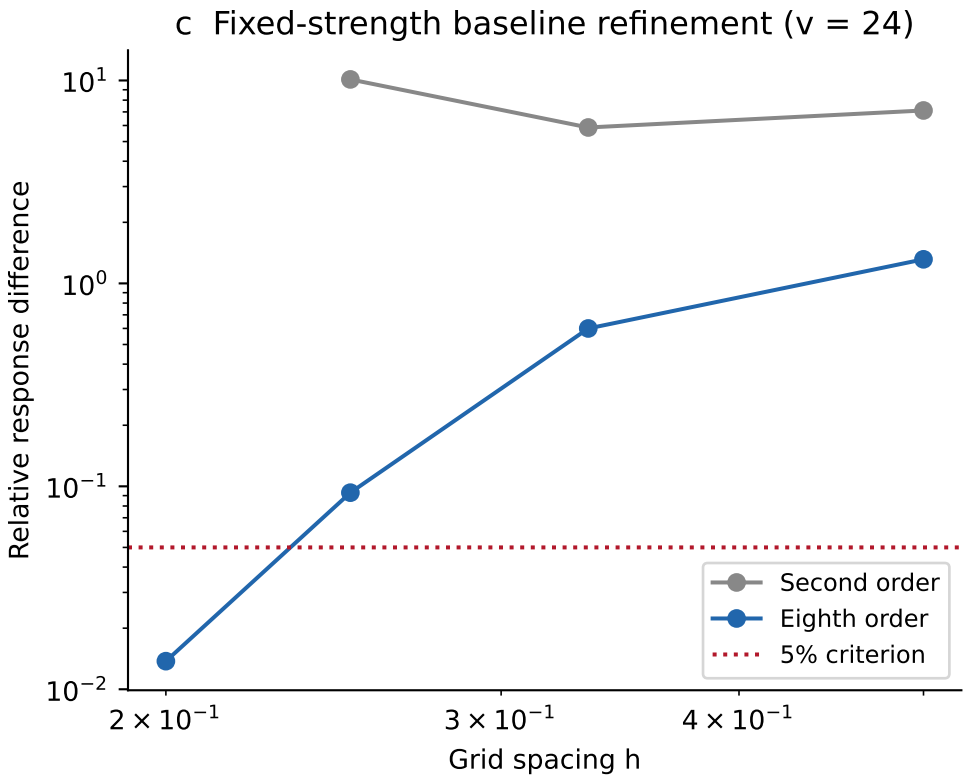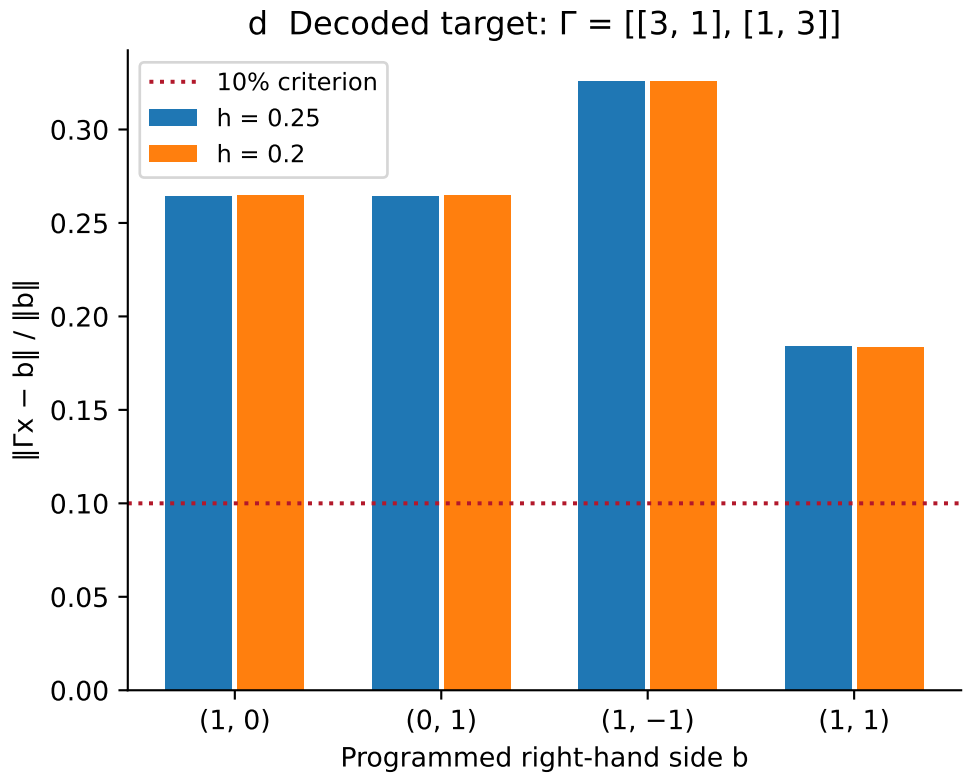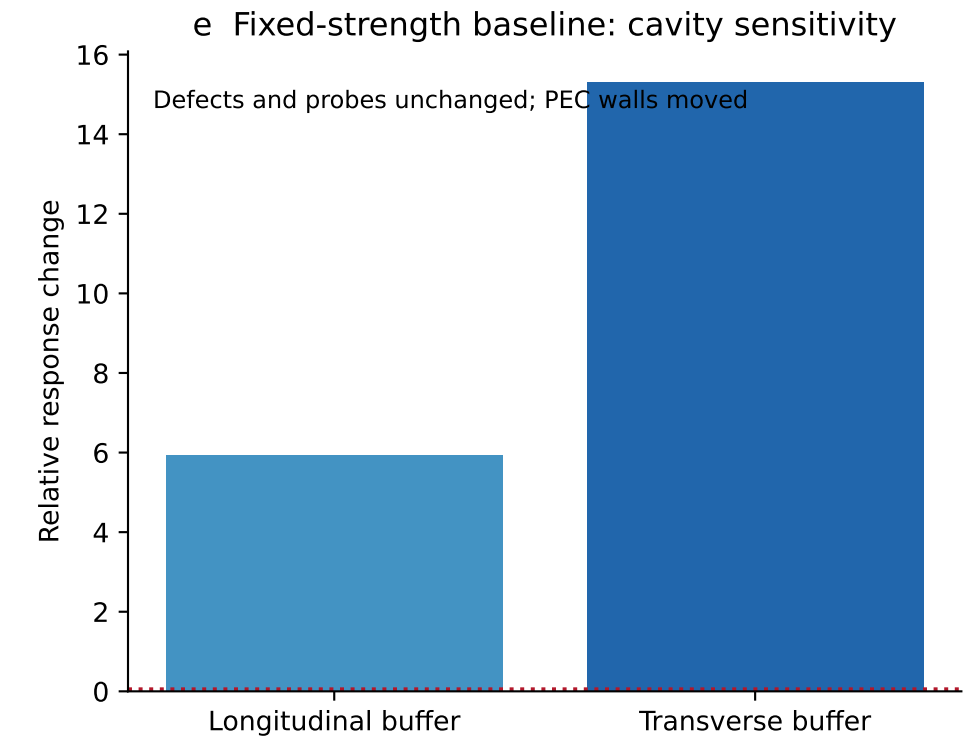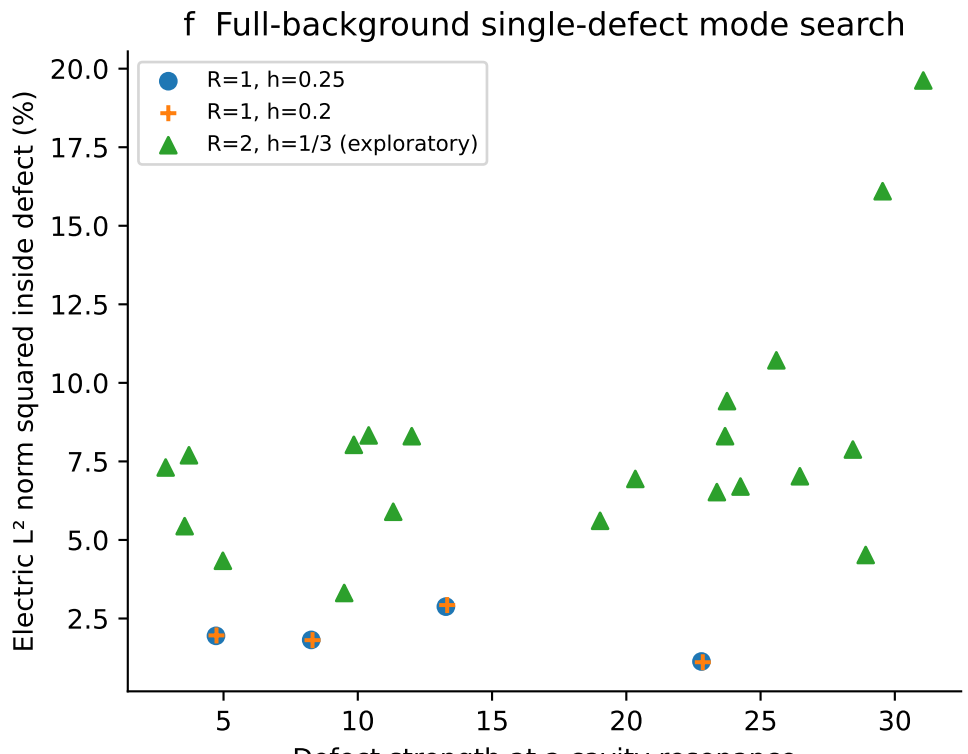The local three-mode approximation fails at the tested compact parameters